THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 4: **Algebraic Identities** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** $x^2 + 4x + 4$	$(a+b)^2 = a^2 + 2ab + b^2$
2	**(d)** $x^2 - 8x + 16$	$(a-b)^2 = a^2 - 2ab + b^2$
3	**(b)** $x^2 - 9$	$(a+b)(a-b) = a^2 - b^2$
4	**(a)** 961	$31 = (30+1)$, apply $(a+b)^2$
5	**(c)** 2499	$(50-1)(50+1) = 2500 - 1$
6	**(a)** $(x+5)(x-5)$	$a^2 - b^2 = (a+b)(a-b)$
7	**(a)** 49	Add first, then square.
8	**(d)** $x^2 + 5x + 6$	$(x+a)(x+b) = x^2 + (a+b)x + ab$
9	**(d)** $x^2 + 6x + 9$	$(a+b)^2 = a^2 + 2ab + b^2$
10	**(a)** $x^2 - 10x + 25$	$(a-b)^2 = a^2 - 2ab + b^2$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Use $(a+b)^2 = a^2 + 2ab + b^2$ (1); $= x^2 + 16x + 64$ (1).	1+1
12	Use $(a+b)(a-b) = a^2 - b^2$ (1); $= x^2 - 64$ (1).	1+1
13	$42 = (40+2)$ (1); $42^2 = 1600 + 160 + 4 = 1764$ (1).	1+1
14	Use $(a+b)^2 = a^2 + 2ab + b^2$ (1); $= x^2 + 20x + 100$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Use $(a+b)^2 = a^2 + 2ab + b^2$ (1); $a = 2x$, $b = 3$ (1); $= 4x^2 + 12x + 9$ (1).	1+1+1
16	Compare with $a^2 + 2ab + b^2$ (1); $a = x$, $b = 3$ (1); $= (x+3)^2$ (1).	1+1+1
17	$= (50+5)(50-5)$ (1); $= 50^2 - 5^2 = 2500 - 25$ (1); $= 2475$ (1).	1+1+1
	OR: Use $(a+b)^2 = a^2 + 2ab + b^2$ (1); $a = 3x$, $b = 1$ (1); $= 9x^2 + 6x + 1$ (1).	1+1+1
18	Compare with $a^2 + 2ab + b^2$ (1); $a = x$, $b = 4$ (1); $= (x+4)^2$ (1).	1+1+1
	OR: $= (50+6)(50-6)$ (1); $= 50^2 - 6^2 = 2500 - 36$ (1); $= 2464$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Use $(a+b)^2 = a^2 + 2ab + b^2$ (1); $(2x)^2 = 4x^2$, $2(2x)(y) = 4xy$ (1); $= 4x^2 + 4xy + y^2$ (1); At $x = y = 1$: 9 (2).	1+1+1+2
	OR: Find two numbers with sum 5 and product 6 (1); The numbers are 2 and 3 (1); $= (x+2)(x+3)$ (1); Check by expanding (2).	1+1+1+2
20	Use $(a+b)^2 = a^2 + 2ab + b^2$ (1); $(3x)^2 = 9x^2$, $2(3x)(2y) = 12xy$ (1); $= 9x^2 + 12xy + 4y^2$ (1); At $x = y = 1$: 25 (2).	1+1+1+2
	OR: Find two numbers with sum 7 and product 12 (1); The numbers are 3 and 4 (1); $= (x+3)(x+4)$ (1); Check by expanding (2).	1+1+1+2

Mistakes to avoid:

- Writing $(a+b)^2 = a^2 + b^2$ and leaving out the middle term $2ab$.
- Using $(a-b)^2 = a^2 - b^2$; this confuses a square of a difference with a difference of squares.
- Sign errors in $(a-b)^3$: the terms alternate in sign, giving $a^3 - 3a^2b + 3ab^2 - b^3$.
- Not squaring the whole term: $(2x+3)^2 = 4x^2 + 12x + 9$, not $2x^2 + 12x + 9$.
- Factorising $x^2 + 5x + 6$ with numbers whose sum or product is wrong; always check that the sum is 5 and the product is 6.
- Not taking out a common factor before applying an identity, e.g. $2x^2 - 18 = 2(x-3)(x+3)$.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 4: **Algebraic Identities** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** $x^2 - 16$	$(a+b)(a-b) = a^2 - b^2$
2	**(d)** 1681	$41 = (40+1)$, apply $(a+b)^2$
3	**(b)** 2496	$(50-2)(50+2) = 2500 - 4$
4	**(d)** $(x+7)(x-7)$	$a^2 - b^2 = (a+b)(a-b)$
5	**(b)** 100	Add first, then square.
6	**(a)** $x^2 + 9x + 20$	$(x+a)(x+b) = x^2 + (a+b)x + ab$
7	**(b)** $x^2 + 10x + 25$	$(a+b)^2 = a^2 + 2ab + b^2$
8	**(a)** $x^2 - 16x + 64$	$(a-b)^2 = a^2 - 2ab + b^2$
9	**(c)** $x^2 - 36$	$(a+b)(a-b) = a^2 - b^2$
10	**(c)** 2704	$52 = (50+2)$, apply $(a+b)^2$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Use $(a+b)(a-b) = a^2 - b^2$ (1); $= x^2 - 144$ (1).	1+1
12	$53 = (50+3)$ (1); $53^2 = 2500 + 300 + 9 = 2809$ (1).	1+1
13	Use $(a+b)^2 = a^2 + 2ab + b^2$ (1); $= x^2 + 24x + 144$ (1).	1+1
14	Use $(a+b)(a-b) = a^2 - b^2$ (1); $= x^2 - 225$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Use $(a+b)^2 = a^2 + 2ab + b^2$ (1); $a = 4x$, $b = 5$ (1); $= 16x^2 + 40x + 25$ (1).	1+1+1
16	Compare with $a^2 + 2ab + b^2$ (1); $a = x$, $b = 5$ (1); $= (x+5)^2$ (1).	1+1+1
17	$= (50+7)(50-7)$ (1); $= 50^2 - 7^2 = 2500 - 49$ (1); $= 2451$ (1).	1+1+1
	OR: Use $(a+b)^2 = a^2 + 2ab + b^2$ (1); $a = 5x$, $b = 2$ (1); $= 25x^2 + 20x + 4$ (1).	1+1+1
18	Compare with $a^2 + 2ab + b^2$ (1); $a = x$, $b = 6$ (1); $= (x+6)^2$ (1).	1+1+1
	OR: $= (50+8)(50-8)$ (1); $= 50^2 - 8^2 = 2500 - 64$ (1); $= 2436$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Use $(a+b)^2 = a^2 + 2ab + b^2$ (1); $(x)^2 = x^2$, $2(x)(4y) = 8xy$ (1); $= x^2 + 8xy + 16y^2$ (1); At $x = y = 1$: 25 (2).	1+1+1+2
	OR: Find two numbers with sum 9 and product 20 (1); The numbers are 4 and 5 (1); $= (x+4)(x+5)$ (1); Check by expanding (2).	1+1+1+2
20	Use $(a+b)^2 = a^2 + 2ab + b^2$ (1); $(2x)^2 = 4x^2$, $2(2x)(5y) = 20xy$ (1); $= 4x^2 + 20xy + 25y^2$ (1); At $x = y = 1$: 49 (2).	1+1+1+2
	OR: Find two numbers with sum 8 and product 15 (1); The numbers are 3 and 5 (1); $= (x+3)(x+5)$ (1); Check by expanding (2).	1+1+1+2

Mistakes to avoid:

- Writing $(a+b)^2 = a^2 + b^2$ and leaving out the middle term $2ab$.
- Using $(a-b)^2 = a^2 - b^2$; this confuses a square of a difference with a difference of squares.
- Sign errors in $(a-b)^3$: the terms alternate in sign, giving $a^3 - 3a^2b + 3ab^2 - b^3$.
- Not squaring the whole term: $(2x+3)^2 = 4x^2 + 12x + 9$, not $2x^2 + 12x + 9$.
- Factorising $x^2 + 5x + 6$ with numbers whose sum or product is wrong; always check that the sum is 5 and the product is 6.
- Not taking out a common factor before applying an identity, e.g. $2x^2 - 18 = 2(x-3)(x+3)$.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 4: **Algebraic Identities** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 2491	$(50-3)(50+3) = 2500-9$
2	**(d)** $(x+9)(x-9)$	$a^2-b^2=(a+b)(a-b)$
3	**(b)** 169	Add first, then square.
4	**(b)** $x^2+10x+21$	$(x+a)(x+b)=x^2+(a+b)x+ab$
5	**(c)** $x^2+12x+36$	$(a+b)^2=a^2+2ab+b^2$
6	**(a)** $x^2-18x+81$	$(a-b)^2=a^2-2ab+b^2$
7	**(b)** x^2-100	$(a+b)(a-b)=a^2-b^2$
8	**(d)** 3721	$61=(60+1)$, apply $(a+b)^2$
9	**(a)** 2484	$(50-4)(50+4)=2500-16$
10	**(d)** $(x+11)(x-11)$	$a^2-b^2=(a+b)(a-b)$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$64=(60+4)$ (1); $64^2=3600+480+16=4096$ (1).	1+1
12	Use $(a+b)^2=a^2+2ab+b^2$ (1); $=x^2+8x+16$ (1).	1+1
13	Use $(a+b)(a-b)=a^2-b^2$ (1); $=x^2-4$ (1).	1+1
14	$82=(80+2)$ (1); $82^2=6400+320+4=6724$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Use $(a+b)^2=a^2+2ab+b^2$ (1); $a=2x$, $b=7$ (1); $=4x^2+28x+49$ (1).	1+1+1
16	Compare with $a^2+2ab+b^2$ (1); $a=x$, $b=7$ (1); $=(x+7)^2$ (1).	1+1+1
17	$=(50+9)(50-9)$ (1); $=50^2-9^2=2500-81$ (1); $=2419$ (1).	1+1+1
	OR: Use $(a+b)^2=a^2+2ab+b^2$ (1); $a=3x$, $b=4$ (1); $=9x^2+24x+16$ (1).	1+1+1
18	Compare with $a^2+2ab+b^2$ (1); $a=x$, $b=9$ (1); $=(x+9)^2$ (1).	1+1+1
	OR: $=(50+10)(50-10)$ (1); $=50^2-10^2=2500-100$ (1); $=2400$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Use $(a+b)^2=a^2+2ab+b^2$ (1); $(4x)^2=16x^2$, $2(4x)(3y)=24xy$ (1); $=16x^2+24xy+9y^2$ (1); At $x=y=1$: 49 (2).	1+1+1+2
	OR: Find two numbers with sum 11 and product 30 (1); The numbers are 5 and 6 (1); $=(x+5)(x+6)$ (1); Check by expanding (2).	1+1+1+2
20	Use $(a+b)^2=a^2+2ab+b^2$ (1); $(3x)^2=9x^2$, $2(3x)(3y)=18xy$ (1); $=9x^2+18xy+9y^2$ (1); At $x=y=1$: 36 (2).	1+1+1+2
	OR: Find two numbers with sum 10 and product 21 (1); The numbers are 3 and 7 (1); $=(x+3)(x+7)$ (1); Check by expanding (2).	1+1+1+2

Mistakes to avoid:

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- Using $(a-b)^2=a^2-b^2$; this confuses a square of a difference with a difference of squares.
- Sign errors in $(a-b)^3$: the terms alternate in sign, giving $a^3-3a^2b+3ab^2-b^3$.
- Not squaring the whole term: $(2x+3)^2=4x^2+12x+9$, not $2x^2+12x+9$.
- Factorising x^2+5x+6 with numbers whose sum or product is wrong; always check that the sum is 5 and the product is 6.
- Not taking out a common factor before applying an identity, e.g. $2x^2-18=2(x-3)(x+3)$.

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